Medical Application & HealthCare Project

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Q1 – Descriptive Analyses and Preprocessing

Task 0 – Data Exploration and Descriptive Analyses

The exploratory analysis of the baseline and longitudinal data was conducted to characterize the study population, assess clinical heterogeneity, and provide the descriptive foundation for subsequent longitudinal and inferential analyses. Baseline variables were grouped into clinically coherent domains to reflect standard practice in observational clinical studies and to facilitate interpretation of treatment effects over time.

Consistency of treatment effects across cycles

Descriptive analyses of longitudinal MMDs show broadly consistent treatment effects across the three consecutive cycles. Median percentage reductions increased from **≈57%** in Cycle 1 to **≈70%** in Cycle 3, reflecting sustained benefit. Interquartile ranges largely overlapped, indicating general consistency of response, while the upward shift of medians suggests maintained—or potentially enhanced—efficacy among patients who continued treatment. Outliers and the decreasing number of patients over cycles due to attrition highlight heterogeneity and the exploratory nature of these observations.

Temporal evolution of baseline MMDs

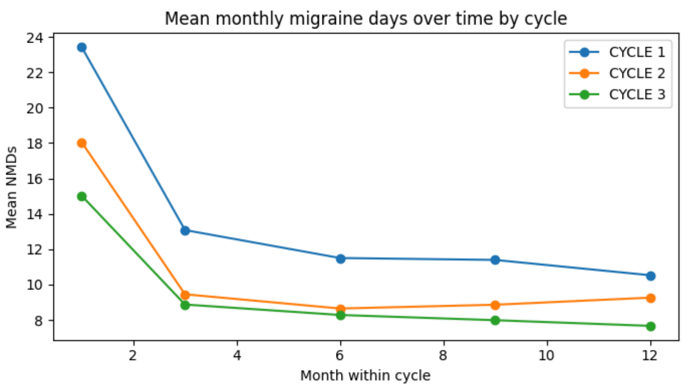
Monthly migraine days decreased sharply during the early months of the first cycle, with reductions from baseline peaking around Month 3 and then plateauing. In Cycles 2 and 3, patients started from lower MMD levels and experienced further modest reductions, consistently below first-cycle values. Parallel improvements were observed for headache intensity and migraine-related disability, indicating aligned benefits in frequency, severity, and functional impact throughout the treatment period.

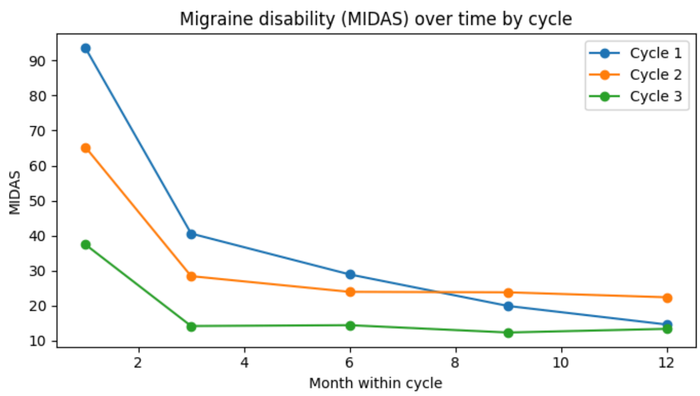
Average Reduction in Monthly Migraine Days Across Three Treatment Cycles – One Year

Across the three consecutive treatment cycles, reductions in monthly migraine days were substantial and generally increased among patients who remained on treatment. In Cycle 1, all **179** patients contributed data, with a mean reduction of **12.4 days** (≈50% from baseline) and a median reduction of **56.7%**. In Cycle 2, **104** patients remained, showing a mean reduction of **14.5 days** (≈59%) and a median of **66.7%.** By Cycle 3, **79** patients contributed, with mean and median reductions of **16.1 days** (≈66%) and 70%, respectively. These results indicate that, despite progressive attrition, patients continuing treatment experienced sustained—and possibly enhanced—benefits over time.

Responder rates

The proportion of patients achieving clinically meaningful responses increased progressively. At the end of Cycle 1, ≈**74%** achieved **≥30% reduction** and **58%** achieved **≥50% reduction**. These rates rose to ≈**86%** and **71%** in Cycle 2 and to ≈**92%** and **82%** in Cycle 3. Across the full treatment period, **72%** and **60%** of patients maintained **≥30%** and **≥50%** reductions, respectively, at their last observation. Despite attrition in later cycles, these data suggest sustained and increasing benefit over time in a clinically heterogeneous population.





Task 1 – Missing Data Imputation

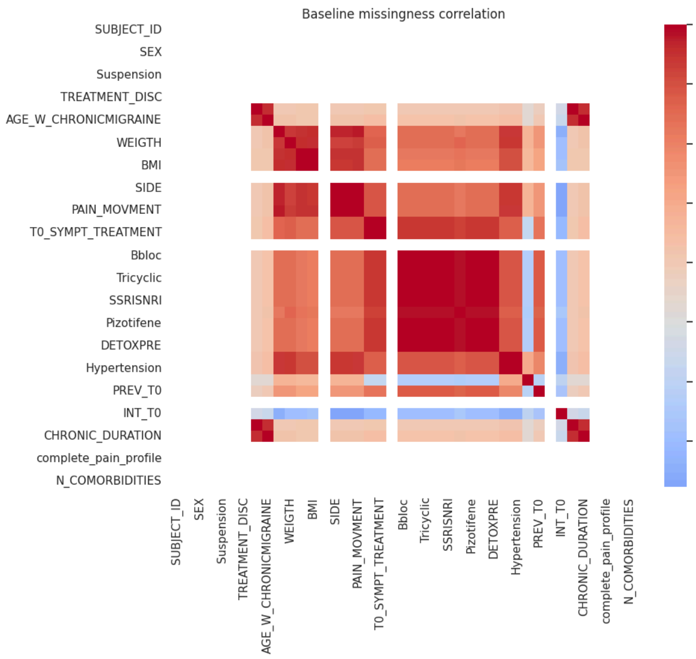
Some groups of variables tended to be missing together (co-missing clusters), reflecting incomplete sections of questionnaires rather than selective exclusion of specific patients. This supports the assumption of MCAR or MAR at baseline, as missingness was unrelated to unobserved values and could be modeled using observed covariates. Consequently, multivariate imputation (MICE) was appropriate, leveraging correlations among variables to impute missing values while preserving the full sample and realistic distributions. In contrast, longitudinal missingness was predominantly dropout-driven and partially monotone, increasing over time and differing across cycles. This justified time-aware imputations such as LOCF, which respect the temporal structure without extrapolating beyond observed data.

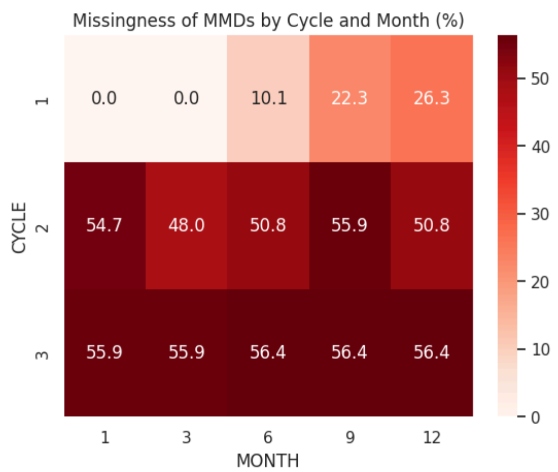
Handling of Missing Data at Baseline

We began by assessing missingness in baseline variables. Using complete-case (CC) analysis, only 46 of 179 subjects were retained, leading to overfitting (~40 predictors, Condition Number = $1{\times}10^{16}$) and unstable estimates (R^2 = 0.946, adj. R^2 = 0.512). Restricting the model to 5 arbitrary chosen predictors improved interpretability, but only 47 subjects remained, with much lower explained variance (R^2 = 0.141).
To avoid losing data, we applied simple imputation, replacing missing continuous values with the median and categorical values with the mode. This preserved the full sample (n = 179) but artificially reduced variability and ignored correlations, distorting distributions. Finally, we implemented MICE with 5 imputations, capturing multivariate relationships and retaining all subjects. Diagnostics confirmed stable means and standard deviations across imputations, and distributions closely matched observed data. Comparing MICE predictions with the reduced CC model yielded RMSE = 1.73, indicating moderate divergence. Overall, MICE provided the most reliable and interpretable estimates, preserving the full sample and properly accounting for uncertainty due to missing data.

Handling of Missing Data at Longitudinal

To handle missing MMDs in the longitudinal dataset, data were first converted to wide format and ordered temporally within each cycle. Three "time-aware" imputation strategies were compared: LOCF (Last Observation Carried Forward), NOCB (Next Observation Carried Backward), and linear interpolation. The original wide dataset contained 432 missing values. Linear interpolation imputed 0 values, reflecting the absence of internal gaps, while NOCB imputed only 28 values (6.5%), often introducing information from later months. In contrast, LOCF recovered 217 missing values (50%), filling most late-cycle dropouts without creating artificial peaks. Analysis of mean MMD trajectories across cycles showed that LOCF closely followed observed trends in early months and only flattened in later months, consistent with expected dropout patterns. The distribution of LOCF-imputed values remained entirely within the observed MMD range, ensuring clinical plausibility. Based on these results, LOCF was chosen for longitudinal imputation, producing a complete long-format dataset suitable for subsequent descriptive and trajectory analyses.





Q2 – Predictors of Response and Discontinuation

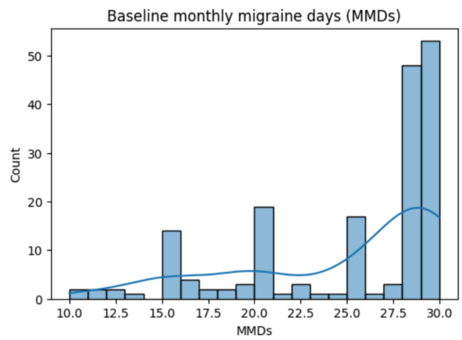
Task 2 – Mixed-Effect Model Analysis

To address the factors influencing treatment outcomes, we implemented a dual-modeling approach: a **Logistic Regression** to identify predictors of discontinuation (dropout) and a **Hierarchical Linear Mixed-Effects Model (LMM)** to analyze the longitudinal reduction of Monthly Migraine Days (MMDs).

Baseline Characteristics Predicting Treatment Response

The longitudinal response was analyzed using an LMM to account for the nested nature of the data (months within patients). The model identified the following baseline predictors:

- **Migraine History**: A higher number of previously failed treatments significantly predicted a greater reduction in MMDs over time (ß = -0.48, $p = 0.018$), highlighting the drug's efficacy in difficult-to-treat populations.
- **Baseline Severity**: Initial frequency was the strongest predictor of absolute MMD levels during follow-up (ß = 0.55, $p < 0.001$); patients starting with a higher burden maintained higher absolute MMDs despite showing improvement.
- **Demographics**: Factors such as age and sex did not show a statistically significant impact on the biological response to treatment ($p > 0.05$).
- **Accountability for Heterogeneity**: The model calculated a **Group Variance of 23.65**, which represents the significant baseline differences in migraine frequency between individual patients. Including this "Random Intercept" was essential to isolate the true treatment effect from the personal starting point of each participant.



Factors Associated with Treatment Discontinuation

The dropout model (Pseudo R-squared = 0.305) identified that clinical complexity, rather than demographics, determines persistence:

- **Baseline Severity and Comorbidities**: Higher baseline Monthly Migraine Days (MMDs) were significantly associated with an increased risk of discontinuation (**OR = 1.10, p = 0.027**). Similarly, a higher number of comorbidities increased the likelihood of treatment suspension (**OR = 1.36, p = 0.043**).
- **Treatment History**: Conversely, a greater number of previously failed preventive treatments acted as a strong stabilizer for adherence, significantly reducing the likelihood of dropout (**OR = 0.48, p < 0.001**). This suggests higher persistence among refractory patients who may have exhausted other therapeutic options.
- **Non-Significant Factors**: Demographics (age, sex), specific migraine diagnosis, and the type of anti-CGRP antibody used did not statistically predict discontinuation (**p > 0.05**)

Analysis Summary

Our findings are supported by a temporal analysis across months and cycles. The primary time effect showed a reduction of **0.87 MMDs per month** ($p < 0.001$). To ensure these results were not biased by informative dropout, we conducted a **Sensitivity Analysis using LOCF imputation**, which confirmed a robust and significant MMD reduction (ß = -0.73, $p < 0.001$) even under conservative assumptions.

Q3 – Migraine Phenotypes and Disease Trajectories

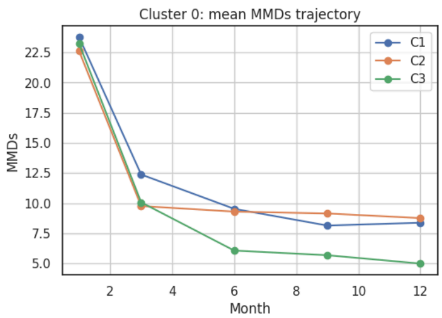
Task 3 – Temporal Data Mining and Electronic Phenotyping

To identify distinct progression patterns, we applied K-Means clustering to high-level temporal features derived from individual patient trajectories—specifically the mean, volatility (standard deviation), and trend slope of MMDs across treatment cycles. This temporal abstraction allowed us to move beyond noisy raw data to identify three interpretable phenotypes.

Emergence of Distinct Patient Phenotypes

Using the Elbow method, k=3 was determined as the optimal number of clusters for patient stratification:

- **Cluster 0 (Stable Responders)**: Patients with a rapid, drastic reduction in MMDs early in Cycle 1, maintaining a consistently low burden (under 10 days) throughout the study.
- **Cluster 1 (Late/Partial Responders)**: Characterized by high early volatility and failure to achieve stable reduction, remaining mostly above 15–20 MMDs.
- **Cluster 2 (High-Burden/Fluctuating)**: Subjects who achieved significant reduction by the end of Cycle 1 and reached a stable, homogeneous state (approx. 7–8 days) in later cycles, despite a complex clinical history.



Temporal Patterns and Prediction of Response

The slope of reduction and volatility proved to be the strongest indicators of long-term outcome. A steep reduction slope in the first cycle was highly predictive of sustained success, while high volatility was a precursor to treatment failure. Crucially, Kruskal-Wallis tests demonstrated that these patterns were not influenced by demographics like age or sex ($p > 0.05$) but were significantly linked to migraine history, specifically the number of previously failed treatments ($p = 0.0025$).

Patient Stratification for Better Outcomes

These phenotypes provide a data-driven framework for personalized management and risk assessment:

- **Persistence Forecasting**: Discontinuation rates varied drastically, with **Cluster 1** showing a **85.9%** suspension rate compared to **0.0%** in **Cluster 0**. This identifies high-volatility patients as requiring intensified monitoring.
- **Refractoriness Management:** Results from Cluster 2 prove that "refractory" patients (those with many prior failures) can still achieve excellent long-term stability. Clinical expectations for these patients should be shifted toward a more gradual, fluctuating trajectory rather than immediate exclusion from therapy.